Reproducing the VCell hybrid methods paper (Schaff et al. 2016) with the native solver and the co-simulation

complete · viva-pde-particle--outcomes · 3 studies · 2026-10-03

Executive summary

Question. Do VCell's native PDE/particle hybrid solver (vcell-fvsolver with embedded Smoldyn) and the process-bigraph co-simulation both reproduce the validation results of Schaff et al. (2016, PLoS Comput Biol 12:e1005236), and where do they differ?

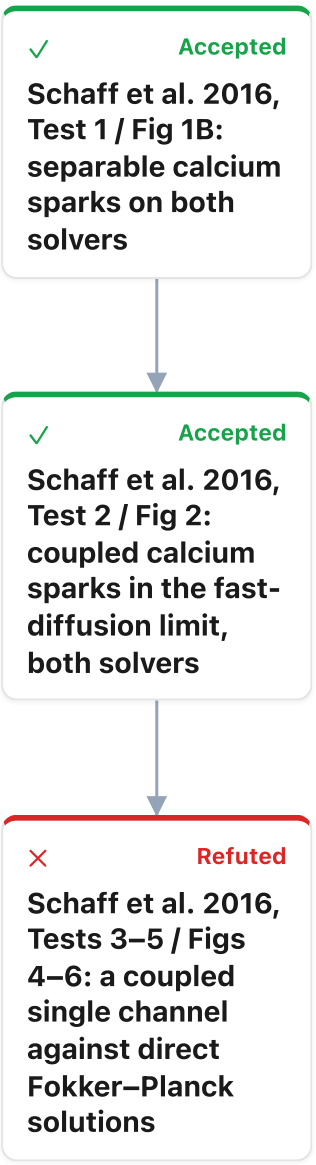
Reproduced. All three Schaff et al. (2016) benchmarks ran on both the native VCell hybrid and the process-bigraph co-simulation, and the two agree to within sampling: • Fig 1B separable calcium sparks — ε falls as N^-1/2 (slopes −0.47 / −0.46), ε≈1.3% at N=128 on both solvers, statistically Δt-dominated; co-sim vs native 0.75× combined error (separable-calcium-sparks). • Fig 2 fast-diffusion open-channel distribution — P(n,t) L2 distance 0.052 / 0.048, both under the paper's 0.1 band (coupled-sparks-fast-diffusion). One knife-edge remains and is reported honestly: single-channel-fokker-planck Test 3 (fast-diffusion) has a co-sim reduced χ² of 2.48, just over the [0, 2] band (native passes at 1.82). Both solvers share the same 0.58% bias, and the Δτ sweep shows it is a first-order time-discretization error (slope −0.51) that vanishes as Δτ→0 — a resolvable discretization effect, not a co-simulation or model defect; Tests 4–5 pass comfortably. Reaching native agreement required two documented VCell encoding fixes (no-op jump process virtualcell/vcell

2158; zero-reactant mass-action leak).

Across **3** studies — **2** supported · **0** qualified · **1** refuted · **0** planned. *Supported* = a computational gate passed under the stated model assumptions (evidence of feasibility, not proof the transition occurred).

The investigation graph

The campaign as a dependency graph — each study a node badged with its code-computed status, wired to the studies it builds on.



Status: ✓ Accepted ◐ Investigating ○ Planned ⊘ Blocked ✕ Refuted

Studies

Expand all | Collapse all

Every study in the campaign as a collapsible card — the question it tests, its verdict, and what it found at a glance. Click a card to expand the full record: model, runs, figures, tests, findings, and the limitations that bound it.

▾ 01 **Schaff et al. 2016, Test 1 / Fig 1B: separable calcium sparks on both**

● ACCEPTED

solvers

Do the native VCell hybrid solver and the process-bigraph co-simulation both reproduce Fig 1B of Schaff et al. (2016)? The error of the trial-mean calcium against its exact expectation should fall as N^-1/2 with the number of trials, and be statistically dominated at small Δt.

Hypothesis & discriminating control

The channels' gating is independent of calcium, so E[U] solves the linear PDE driven by the exact open probability p(t) = k_on/(k_on+k_off)·(1 – e^-(k_on+k_off)t). Both solvers' trial means converge to it with ε ∝ N^-1/2. At N = 128 the statistical error dominates the Δt error for Δt ≤ 0.02 s. The two solvers differ only by sampling.

MODEL EXPERIMENT Model (Eq. 1 of the paper): $\partial U/\partial t = D\nabla^2 U + J\cdot\Sigma_i\, \xi_i\, \delta(r - r_i) - V_p(U - U_0)$, no-flux boundaries. D = 1 μm²/s, J = 10 μM·μm³/s, U_0 = 0.1 μM, V_p = 1 s^{-1}, k_on = 1 s^{-1}, k_off = 5 s^{-1}, 24 channels starting closed, T = 5 s. Geometry: [0, 10.1]×[0, 2.1]×[0, 0.5] μm³, Δx = Δy = 0.1. Runs: 128 trials per solver at Δt ∈ {0.02, 0.01, 0.005} s. ε(N) is taken from nested subsets at Δt = 0.005, and ε is the RMS over space and time of (trial mean – E[U]), relative to max(E[U] – U_0).

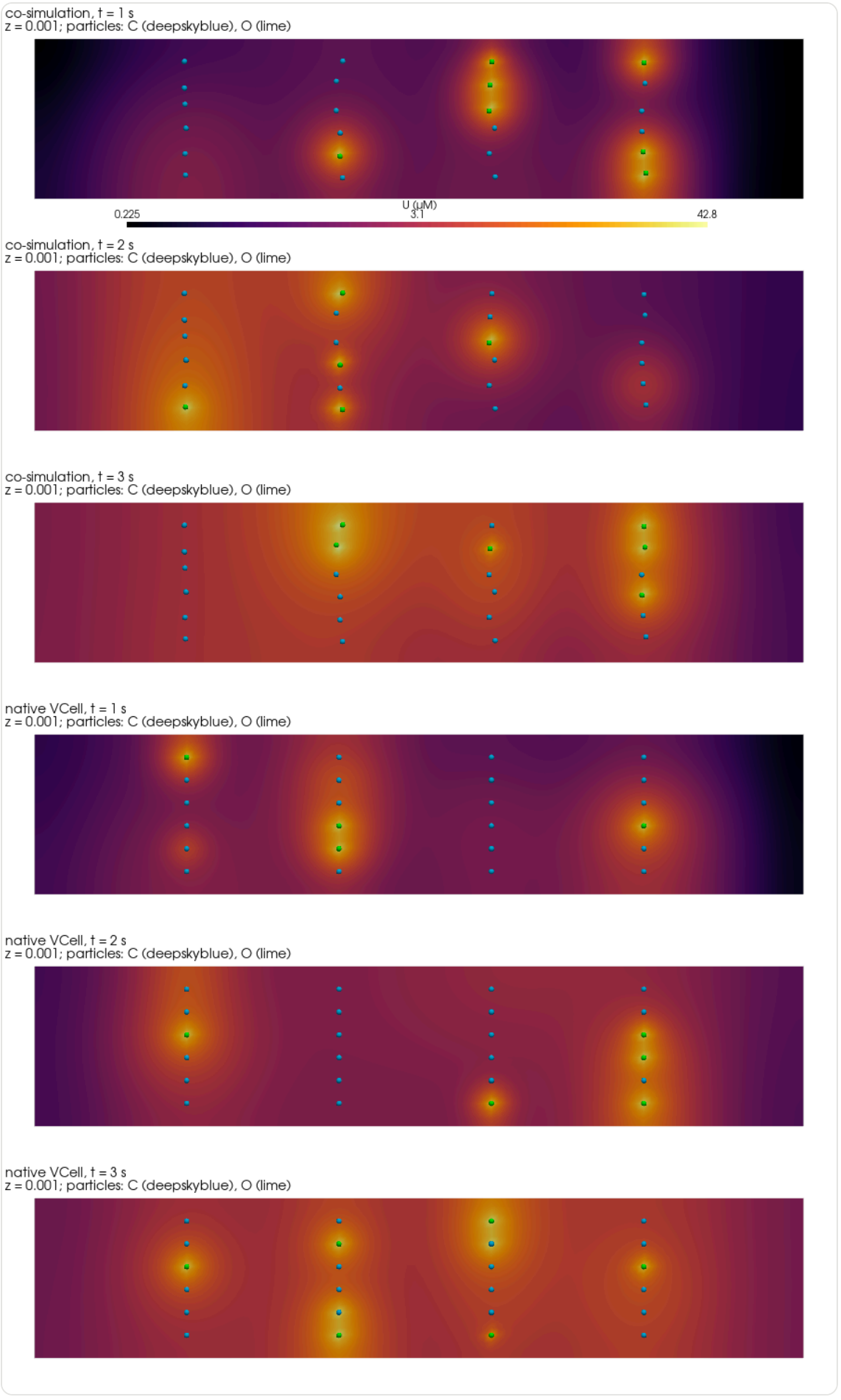
OBSERVABLE `c1_cosim_eps_slope, c1_native_eps_slope, c1_cosim_eps_n128, c1_native_eps_n128, c1_cosim_vs_native_eps_ratio…`

Model & key assumptions

Methods · Simulations

Model	`calcium_sparks`
Sims	1 run
Tests	✓✓✓✓ 4/4 passed

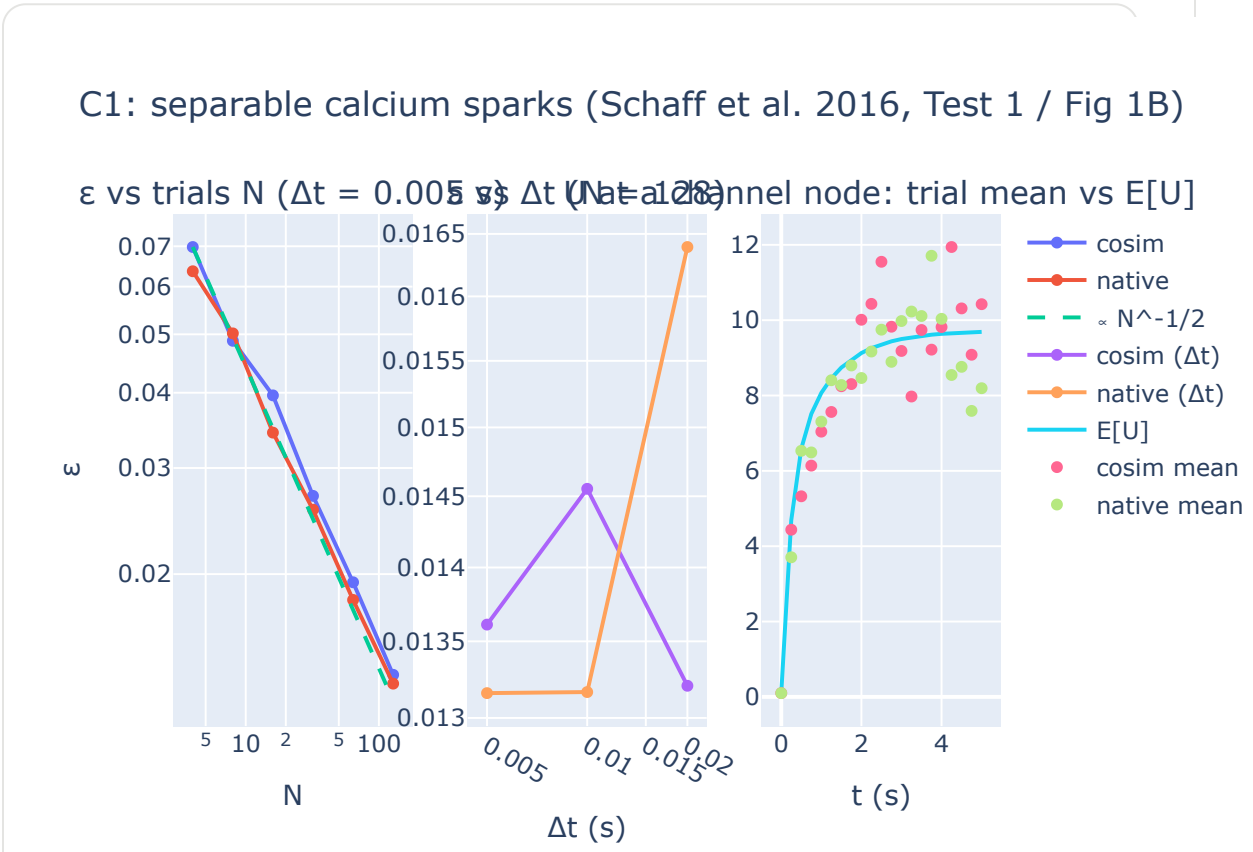
Visualizations



sparks-3d

↪ sparks-3d-movie — too large to inline

CALCIUM SPARKS FIG1B



↪ Sparks 3d interactive — too large to inline

Finding

F-01 · confirms ● Observed

Both solvers reproduce Fig 1B's Monte-Carlo convergence: the log-log slope of ε vs N is −0.47 (co-simulation) and −0.46 (native VCell hybrid), both within sampling of the theoretical −1/2. Schaff 2016

from test: cosim-error-falls-as-inverse-sqrt-n `range` BAND Schaff 2016

runs: `default`

F-02 · confirms ● Observed

At N=128, Δt=0.005 s the error is small and essentially solver-independent: ε=1.37% (co-sim) vs 1.32% (native), and ε is flat across Δt=0.02–0.005 s (1.3–1.6%) — i.e. statistically dominated, exactly as the paper reports. Schaff 2016

from test: cosim-error-small-at-128-trials `< 0.03` BAND Schaff 2016

runs: `default`

F-03 · confirms ● Observed

The co-simulation and native solvers agree to within sampling: their trial-mean difference is 0.75× the combined statistical error (<1.5), so decomposing the hybrid solver into process-bigraph processes does not change the answer. Reaching this required two documented VCell fixes in the native path (no-op jump process virtualcell/vcell#2158; zero-reactant mass-action leak). Schaff 2016

from test: solvers-differ-only-by-sampling < 1.5 BAND Schaff 2016

runs: default

Verdict

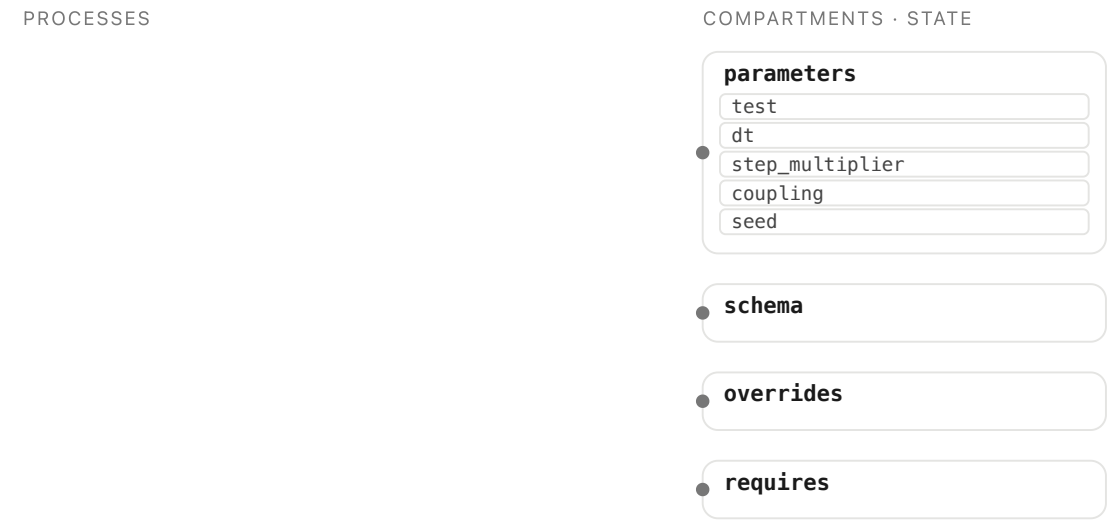
PASSED

Gate cleared · Accepted

Reproduced on both solvers (results/metrics.json, 2026-10-03; run in the dev pixi env): - ε vs N slopes: −0.46 for both solvers, against −1/2; - ε at N = 128: 1.37% (co-sim) and 1.32% (native); - ε is flat across Δt = 0.02–0.005 s (1.3–1.6%), i.e. statistically dominated, as in the paper; - co-sim vs native difference: 0.75× the combined statistical error. Two VCell findings for the native path, both handled in vcell_native.to_biomodel and in post-processing: - **No-op jump process** (virtualcell/vcell
2158): a particle catalysing a continuous source produces a no-op Smoldyn reaction that suppresses channel closing. - **Zero-reactant mass action:** VCell keeps only −Kr·Π(products) and discards Kf, so the constant leak $V_p \cdot U_0$ vanished. It is now written as General kinetics. An earlier run without this fix gave a native ε(128) of 1.6%.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of viva_pde_particle.composites.examples.calcium_sparks — 0 processes wired to 4 compartments. ● writes · ● reads.

viva_pde_particle.composites.examples.calcium_sparks

Simulation run

default

cosim-error-falls-as-inverse-sqrt-n authored: PASS

native-error-falls-as-inverse-sqrt-n authored: PASS

cosim-error-small-at-128-trials authored: PASS

`solvers-differ-only-by-sampling` authored: PASS

Reproduce: `vwb run study separable-calcium-sparks`
dt-sweep: `vwb run study separable-calcium-sparks --variant dt-sweep`
native-vcell-arm: `vwb run study separable-calcium-sparks --variant native-vcell-arm`

Tests

Log-log slope of ε vs N for the co-simulation (Δt = 0.005 s). `cosim-error-falls-as-inverse-sqrt-n · primary · derived_scalar` Schaff et al. 2016, Fig 1B; Monte Carlo error ∝ N^-1/2 when statistical error dominates.	`range undefined`	pending
Log-log slope of ε vs N for the native VCell hybrid (Δt = 0.005 s). `native-error-falls-as-inverse-sqrt-n · primary · derived_scalar` Schaff et al. 2016, Fig 1B.	`range undefined`	pending
ε of the co-simulation at N = 128, Δt = 0.005 s. `cosim-error-small-at-128-trials · secondary · derived_scalar` About 4% at N = 16 scales to about 1.4% at N = 128 under N^-1/2.	`< 0.03`	pending
ε between the co-simulation and native trial means, divided by the combined statistical error sqrt(ε_cosim² + ε_native²), at N = 128 and Δt = 0.005 s. `solvers-differ-only-by-sampling · primary · derived_scalar` Independent ensembles of the same expectation; the ratio is about 1 when only sampling differs.	`< 1.5`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

▾ 02 Schaff et al. 2016, Test 2 / Fig 2: coupled calcium sparks in the fast-diffusion limit, both solvers

● ACCEPTED

With channel opening coupled to local calcium ($k_{on} \cdot U(r_i)/U_0$) and fast diffusion ($D = 1000\ \mu m^2/s$), do both hybrid solvers reproduce the distributions of calcium and open-

channel number given by the exact well-mixed process, as in Fig 2 of the paper?

Hypothesis & discriminating control

At D = 1000 the cell is effectively well mixed. The spatial hybrid's volume-averaged U and open count n then follow the well-mixed piecewise-deterministic process, which the paper sampled with the Gibson–Bruck method; here it is sampled exactly by event-driven simulation (viva_pde_particle.benchmarks.pdmp).

MODEL EXPERIMENT Parameters as in Test 1, except k_on = 0.1 s^{-1} (opening at $k_{on} \cdot U/U_0$), D = 1000 $\mu m^2/s$ and Δt = 0.2 ms; 500 trials per solver. The reference is 20,000 exact realizations of the well-mixed process with V = 10.1×2.1×0.5 μm^3. At t = 1, 2, 3 s the comparison uses: - mean |z| of U and n against the reference; - the L2 distance between histograms, as in the paper; - two-sample KS tests, against the reference and between the solvers.

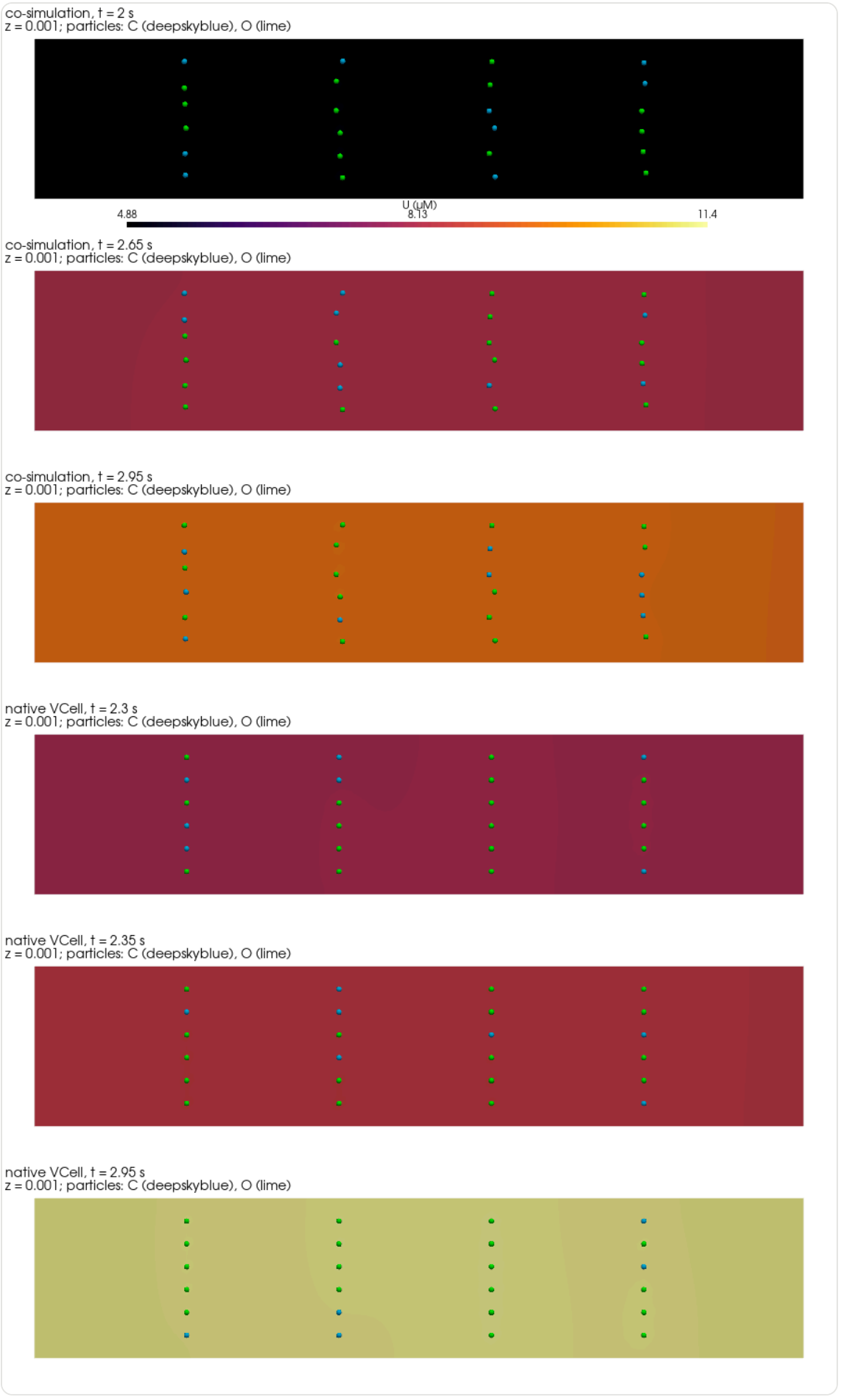
OBSERVABLE `c2_cosim_max_mean_z, c2_native_max_mean_z, c2_cosim_max_l2_n, c2_native_max_l2_n, c2_cosim_vs_native_min_ks_pvalue…`

Model & key assumptions

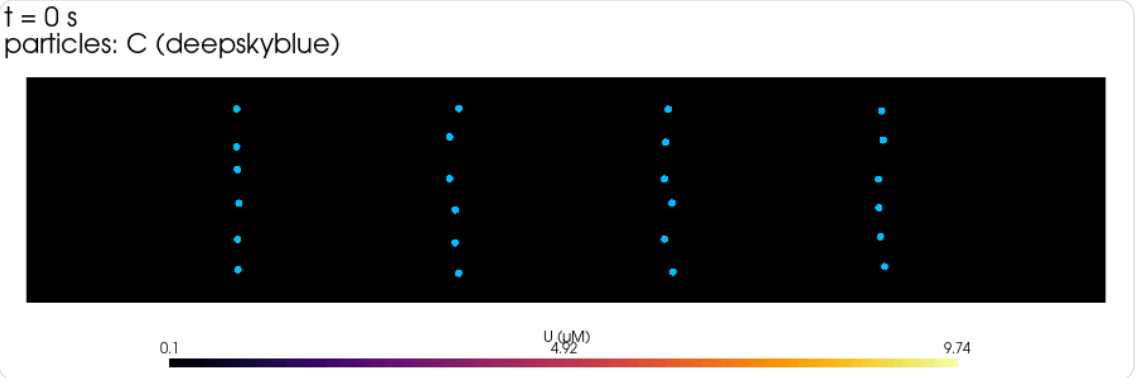
Methods · Simulations

Model	`calcium_sparks`
Sims	1 run
Tests	✓✓✓✓✓ 5/5 passed

Visualizations

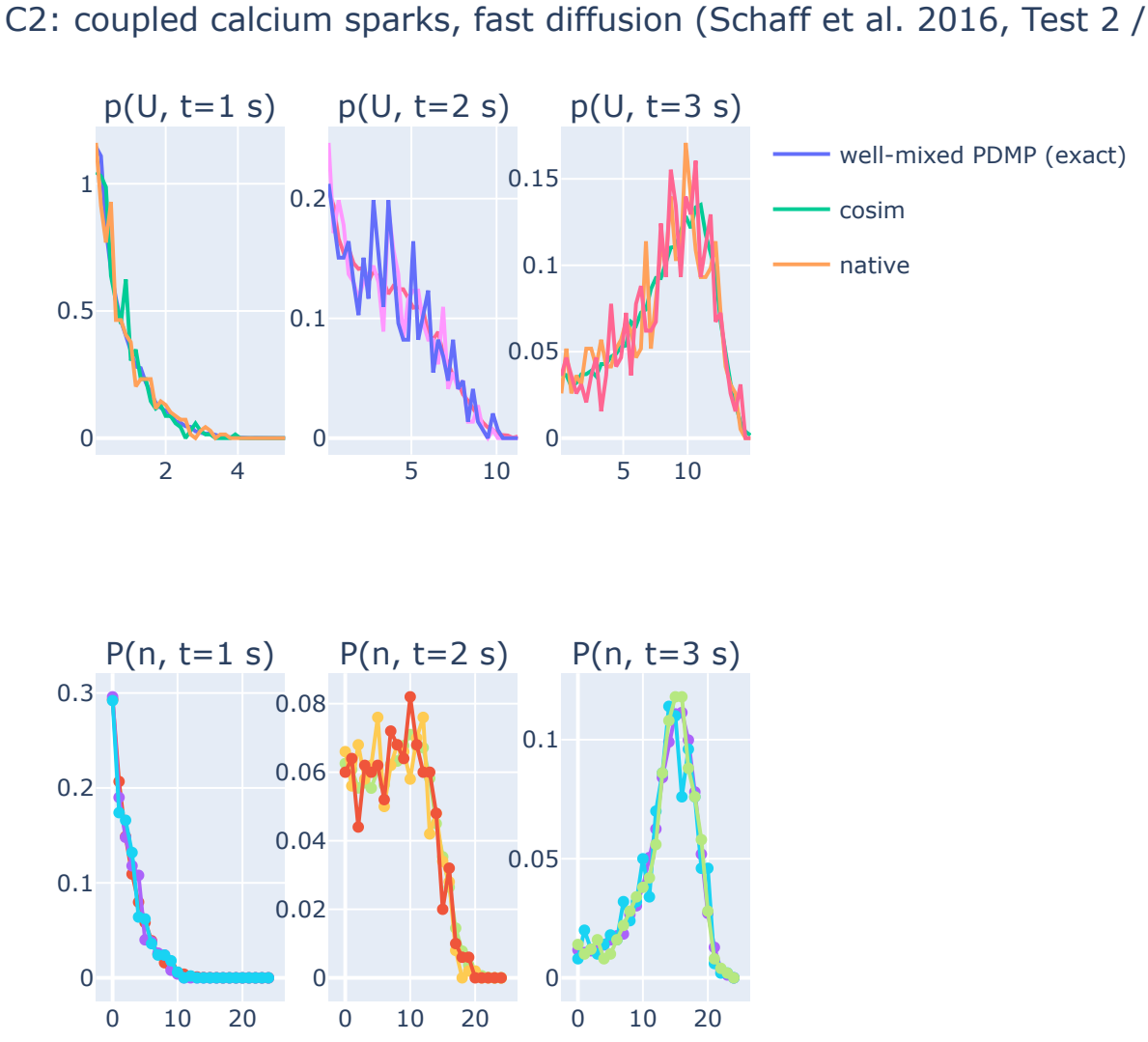


coupled-sparks-3d



coupled-sparks-3d-movie

CALCIUM SPARKS FIG2



↪ Coupled sparks 3d interactive — too large to inline

Finding

F-01 · confirms ● Observed

In the fast-diffusion limit (D = 1000 μm²/s) both hybrid solvers match the exact well-mixed process in mean: max |z| over t∈{1,2,3}s and {U,n} is 1.57 (co-sim) and 0.64 (native VCell), both well under the |z| < 4 threshold. Schaff 2016

from test: cosim-matches-well-mixed-means < 4 BAND Schaff 2016

runs: `default`

F-02 · confirms ● Observed

The open-channel-number distribution P(n, t) reproduces the paper's Fig 2 metric to the 500-trial sampling floor: max-over-t L2 distance is 0.052 (co-sim) and 0.048 (native), both below the 0.1 band (the paper's 10,000-realization value was ~0.002–0.003). Schaff 2016

from test: cosim-channel-distribution-l2 < `0.1` BAND Schaff 2016

runs: `default`

F-03 · confirms ● Observed

The co-simulation and native VCell solvers agree with each other: their minimum KS p-value over t and {U,n} is 0.035 (> 0.001), so decomposing the coupled fast-diffusion hybrid into process-bigraph processes is answer-preserving. Schaff 2016

from test: solvers-agree > `0.001` BAND Schaff 2016

runs: `default`

F-04 · partial ● Observed

Open observation (not a gated test): both solvers show the same small marginal minimum KS p against the well-mixed reference — $7.70 \cdot 10^{-4}$ (co-sim) and $6.87 \cdot 10^{-4}$ (native). The effect is shared across solvers, so it is not a co-simulation artifact; a plausible cause is residual spatial structure (local U at an open channel slightly exceeds the volume mean). Schaff 2016

from test: solvers-agree > `0.001` BAND Schaff 2016

runs: `default`

Verdict

● PASSED **Gate cleared** · Accepted

Results (results/metrics.json, 2026-10-03; 500 trials per solver): - both solvers match the exact well-mixed process in mean: max |z| = 1.57 (co-sim) and 0.64 (native); - P(n, t) L2 distance: 0.052 (co-sim) and 0.048 (native), at the 500-trial sampling floor; - the solvers agree with each other (min KS p = 0.035). Open observation: both solvers show the same marginal minimum KS p against the well-mixed reference (about $7 \cdot 10^{-4}$), so the effect is shared and not a solver artifact. A plausible cause is residual spatial structure: at D = 1000 the local U at an open channel slightly exceeds the volume mean, raising its opening rate. Larger ensembles, or a D → ∞ check, would settle it (follow-up).

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.calcium_sparks` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.calcium_sparks`

Simulation run

default

`cosim-matches-well-mixed-means` authored: PASS

`native-matches-well-mixed-means` authored: PASS

`cosim-channel-distribution-l2` authored: PASS

`native-channel-distribution-l2` authored: PASS

`solvers-agree` authored: PASS

Reproduce: `vwb run study coupled-sparks-fast-diffusion`

cosim-arm: `vwb run study coupled-sparks-fast-diffusion --variant cosim-arm`

native-vcell-arm: `vwb run study coupled-sparks-fast-diffusion --variant native-vcell-arm`

well-mixed-reference: `vwb run study coupled-sparks-fast-diffusion --variant well-mixed-reference`

Tests

Max over t ∈ {1, 2, 3} s and over {U, n} of |z| of the co-simulation mean vs the exact well-mixed process. < 4 pending

`cosim-matches-well-mixed-means · primary · derived_scalar`

Fast-diffusion limit; 500 trials against 20,000 exact realizations.

Max |z| of the native VCell hybrid mean vs the exact well-mixed process. < 4 pending

`native-matches-well-mixed-means · primary · derived_scalar`

As above.

Max over t of the L2 distance between the P(n, t) histograms of the co-simulation and the reference.

< 0.1 pending

`cosim-channel-distribution-l2 · primary · derived_scalar`

The paper's metric (Fig 2, 10,000 realizations gave about 0.002–0.003). With 500 trials the sampling floor is about 0.05.

Max over t of the L2 distance between the P(n, t) histograms of native VCell and the reference.

< 0.1 pending

`native-channel-distribution-l2 · primary · derived_scalar`

As above.

Min KS p-value over t and {U, n}, co-simulation vs native.

> 0.001 pending

`solvers-agree · primary · derived_scalar`

The two solvers should agree; 6 comparisons with ties in n make KS conservative.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

03 Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions

● REFUTED

With a single channel whose opening is coupled to calcium (rate aβ(ρ(0) + 1)), does the hybrid co-simulation reproduce the probability densities of calcium given by direct, Monte-Carlo-free solutions of the Fokker–Planck equations? Two regimes: fast diffusion (well mixed, Test 3) and finite diffusion on coarse grids (Tests 4–5), as in Figs 4–6 of the paper.

Hypothesis & discriminating control

The hybrid samples the same piecewise-deterministic process the Fokker–Planck equations describe, so its histograms should match them to within sampling noise (reduced $\chi^2 \approx 1$). The paper reports L2 differences of 1–3% of the maximum.

MODEL EXPERIMENT Dimensionless model: $\partial\tau\rho = d\cdot\partial^2_x\rho + a\cdot\delta(x)\cdot\xi - \rho$. The channel opens at $\alpha\beta(\rho(0) + 1)$ and closes at α; it starts closed, with $\rho = 0$. - **Test 3:** $\alpha = \beta = 1$, $a = 24$, $d = 10^4$, 3D cube of side 1 ($\Delta x = 0.1$, channel at the centre), $\Delta\tau = 0.002$. - Compared: $p(\bar{\rho})$ at $\tau = 30$. - Reference: Eq 4, solved in time and checked against its exact steady state. - Ensembles: 10,000 co-sim trials, and the same on native VCell. - **Test 4:** $\alpha = 10$, $\beta = 1$, $a = 20/3$, $d = 1$, two cells with $\Delta x = 2$, $\Delta\tau = 10^{-4}$. - Compared: $p(\rho_i)$ at $\tau = 1$. - Reference: the S2-Text master

equation (Eq T2.1) at $\Delta\rho = 2.5, 1.2$ and 0.6×10^{-3}, extrapolated to $\Delta\rho \to 0$. - Ensembles: 12,500 trials. - **Test 5:** $\alpha = 20$, $\beta = 0.5$, $a = 20$, $d = 50$, three cells. - Reference: four $\Delta\rho$, extrapolated as in the paper. - Ensembles: 12,500 trials at $\Delta\tau = 10^{-4}$. Densities are 20-bin histograms on $[0, \rho_{\max(i)}]$ (40 bins on [0, 24] for Test 3), compared with the reference averaged over the same bins. Reported: the relative L2 (RMS over bins / reference maximum), the absolute L2, and the reduced χ^2 against multinomial sampling noise.

OBSERVABLE `c3_test3_cosim_chi2, c3_test3_native_chi2, c3_test4_i0_cosim_chi2, c3_test5_i0_cosim_chi2, c3_dtau_mean_bias_offset…`

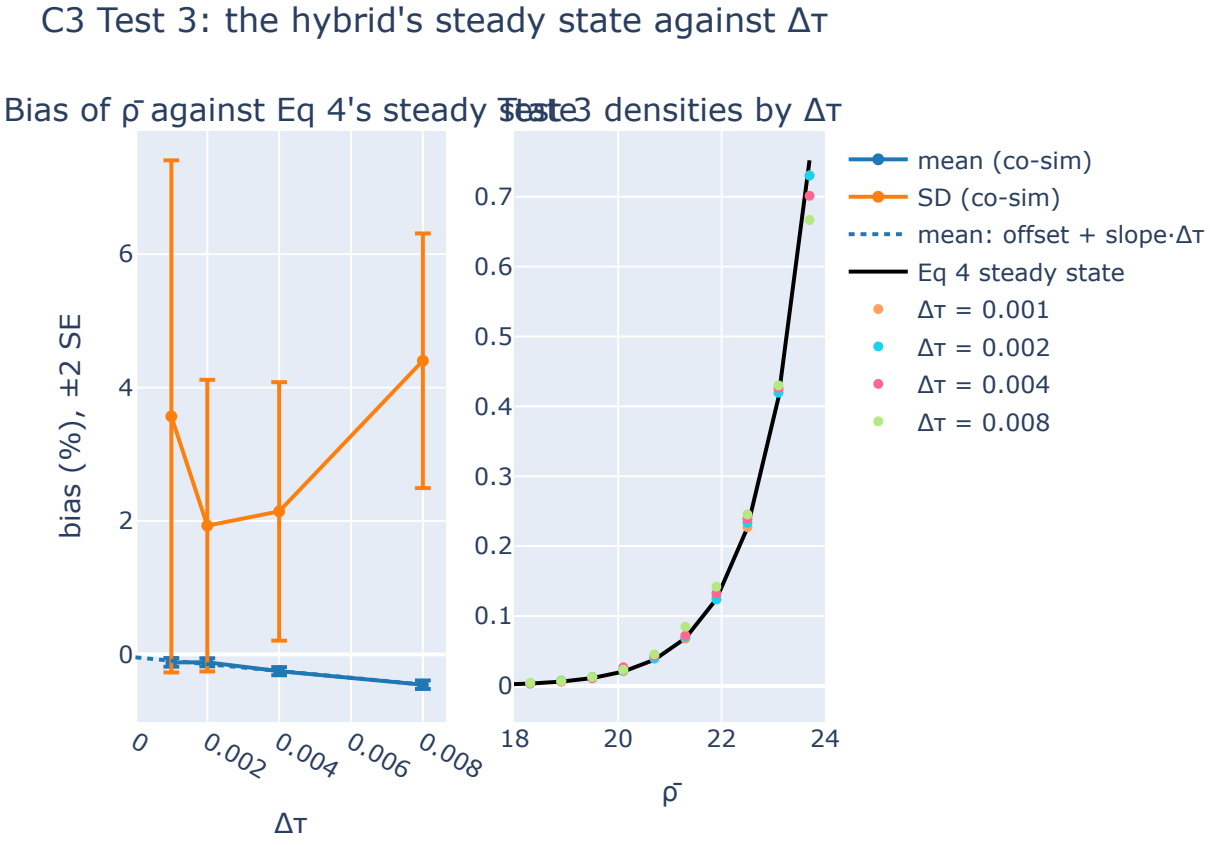
Model & key assumptions

Methods · Simulations

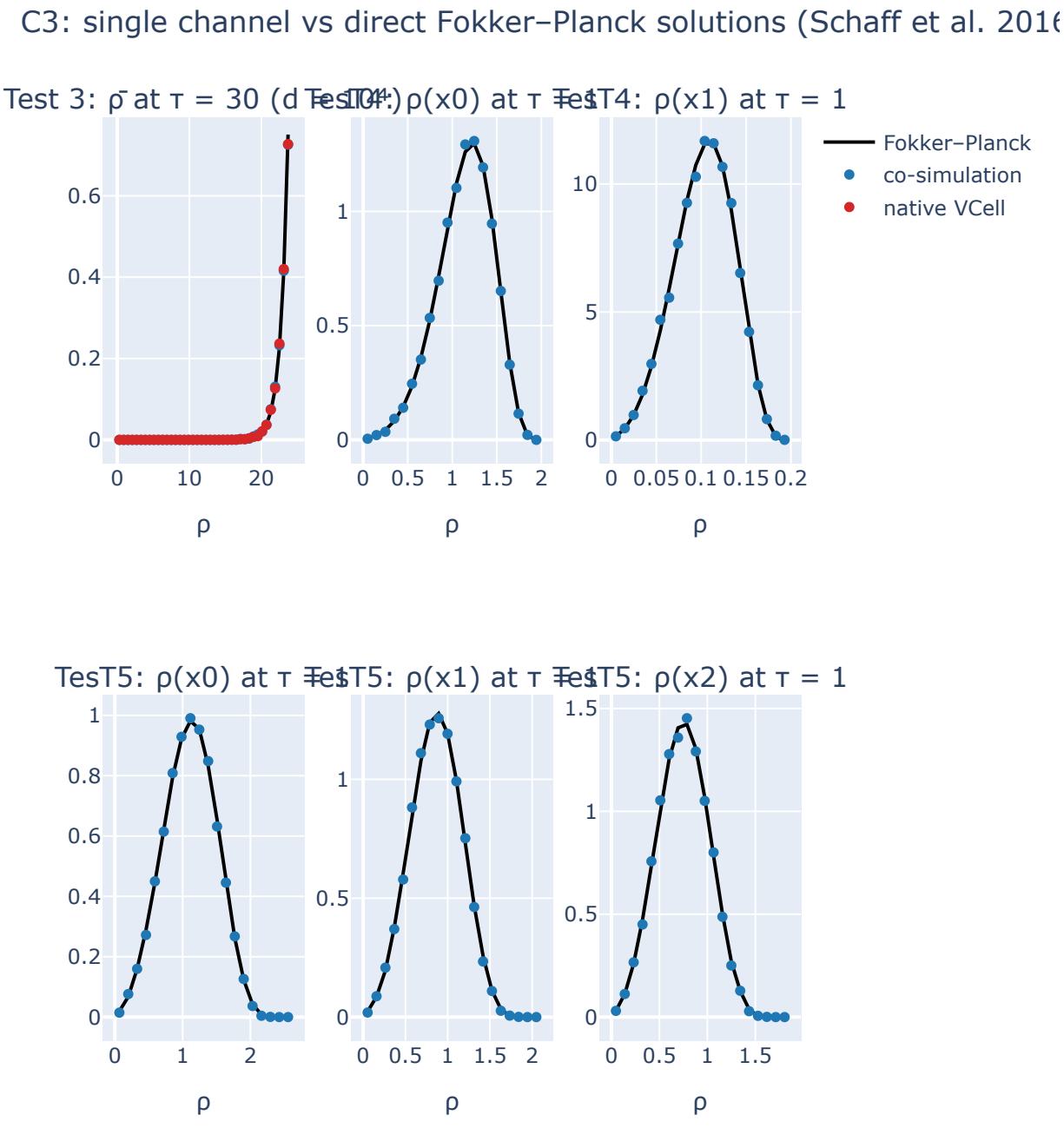
Model	`calcium_sparks`
Sims	1 run
Tests	✓✓✓✓✕ 4/5 passed

Visualizations

DTAU



FOKKER PLANCK



Finding

F-01 · confirms · Observed

Finite diffusion (Tests 4–5): the co-simulation matches the direct Fokker–Planck solutions to within sampling noise in every cell — reduced $\chi^2 = 0.54$ (Test 4, x_0), 0.80 (Test 4, x_1) and 0.80 / 0.89 / 0.86 (Test 5, $x_0/x_1/x_2$), all inside [0, 2]. Relative L2 differences (1.0–1.6%) are smaller than the paper's (1.9% / 3.3% for Test 4).

Schaff 2016

from test: test4-cosim-matches-fokker-planck `range` BAND Schaff 2016

runs: `default`

F-02 · partial · Hypothetical

DIVERGENCE (failed primary): in the fast-diffusion regime (Test 3) the co-simulation's reduced χ^2 against Eq 4 is 2.48, just over the [0, 2] band, so test3-cosim-matches-fokker-planck FAILS. The native VCell arm sits just under at 1.82 (passes). Both solvers share the same small 0.58% relative-L2 bias (about twice the 0.30% sampling noise), so the gap belongs to the spatial hybrid at $\Delta\tau = 0.002$, not to the co-simulation.

Schaff 2016

from test: test3-cosim-matches-fokker-planck `range` BAND Schaff 2016

runs: `default`

F-03 · contradicts ● Hypothetical

The Test 3 gap is a first-order time-discretization error, not a model error: the mean-bias fit offset + slope·$\Delta\tau$ gives a slope of -0.51 ± 0.06 and an offset of $-0.044\% \pm 0.027\%$, consistent with zero (passes $[-0.001, 0.001]$). Native VCell at $\Delta\tau = 0.002$ matches the fit's prediction, confirming the bias scales with $\Delta\tau$ and vanishes as $\Delta\tau \to 0$.

Schaff 2016

from test: test3-bias-is-first-order-in-dtau `range` BAND Schaff 2016

runs: `default`

Verdict

● REFUTED

Gate refused · Reported

Results (results/metrics.json, 2026-10-05; relative L2 = RMS over bins / reference maximum): | test | solver, trials | rel. L2 | sampling noise | reduced χ^2 | paper | |---|---|---|---|---|---| | 3 ($d = 10^4$), $\bar{\rho}$ at $\tau = 30$ | co-sim, 10,000 | 0.58% | 0.30% | 2.48 | abs. L2 0.0137 (ours 0.021) | | 3 | native VCell, 10,000 | 0.58% | 0.30% | 1.82 | | | 4 ($d = 1$), $\rho(x_0)$ / $\rho(x_1)$ at $\tau = 1$ | co-sim, 12,500 | 1.0% / 1.6% | 1.5% / 1.7% | 0.54 / 0.80 | 1.9% / 3.3% | | 5 ($d = 50$), $\rho(x_0)$ / $\rho(x_1)$ / $\rho(x_2)$ | co-sim, 12,500 | 0.9% / 1.2% / 1.4% | ≈ 1.4% | 0.80 / 0.89 / 0.86 | 1.3% / 1.3% / 1.5% | - **Finite diffusion (Tests 4–5):** the co-simulation matches the functional Fokker–Planck solutions to within sampling noise in every cell (reduced χ^2 0.54–0.89), with smaller differences than the paper reports. - Extrapolating the reference to $\Delta\rho \to 0$ matters: it shifts the Test 5 reference by about 4% of its maximum and Test 4's cell 1 by 1.7%. These are the size of the discrepancies the paper attributes to under-resolved $\Delta\rho$. - **Fast diffusion (Test 3):** the reference itself is exact. Eq 4 at $\tau = 30$ agrees with its closed-form steady state to 0.03%. - Both hybrid solvers differ from it by the same 0.58%, about twice the sampling noise. Native VCell's reduced χ^2 (1.82) sits just under the threshold, and the co-sim's (2.48) just over it. - Since the two solvers share the same small bias, it belongs to the spatial hybrid at $\Delta\tau = 0.002$, not to the co-simulation. - **$\Delta\tau$ check (2026-10-08; `dtau` stage, viz/dtau.html):** the Test 3 gap is a first-order time-discretization error. - 4,000 co-sim trials per $\Delta\tau$, each sampled at $\tau = 10, 15, \ldots, 30$ (stationary; lag-5 correlation ≈ 0.01). SEs are bootstrapped over trials. | $\Delta\tau$ | mean bias | SD bias | reduced χ^2 | |---|-

--|---|---| | 0.008 | −0.45% ± 0.03% | +4.4% ± 1.0% | 28.0 | | 0.004 | −0.25% ± 0.03% | +2.1% ± 1.0% | 10.6 | | 0.002 | −0.12% ± 0.03% | +1.9% ± 1.1% | 2.28 | | 0.001 | −0.12% ± 0.03% | +3.6% ± 1.9% | 2.74 | - **Mean:** the fit offset + slope·Δτ gives a slope of −0.51 ± 0.06 (−0.10% at Δτ = 0.002) and an offset of −0.04% ± 0.03%, consistent with zero (reduced χ^2 0.71). Native VCell at Δτ = 0.002 (−0.10% ± 0.04%) matches the slope's prediction. - **SD:** only Δτ = 0.008 is clearly off. The others are within about 2σ. The skewed density (∝ e^ρ) makes the SD's sampling error 2–4× the normal-theory value. - The reduced χ^2 stays above 2 down to Δτ = 0.001. With 20,000 samples, the histogram resolves a 0.1% shift of a density that is steep near ρ = a.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

Model — composite internals



Static bigraph of `viva_pde_particle.composites.examples.calcium_sparks` — 0 processes wired to 4 compartments. ● writes · ● reads.

`viva_pde_particle.composites.examples.calcium_sparks`

Simulation run

default

`test3-cosim-matches-fokker-planck` authored: FAIL

`test3-native-matches-fokker-planck` authored: PASS

`test4-cosim-matches-fokker-planck` authored: PASS

`test5-cosim-matches-fokker-planck` authored: PASS

`test3-bias-is-first-order-in-dtau` authored: PASS

Reproduce: `vwb run study single-channel-fokker-planck`
test3-fast-diffusion: `vwb run study single-channel-fokker-planck --variant test3-fast-diffusion`
test4-finite-diffusion: `vwb run study single-channel-fokker-planck --variant test4-finite-diffusion`

test5-finite-diffusion: `vwb run study single-channel-fokker-planck --variant test5-finite-diffusion`
dtau-sweep: `vwb run study single-channel-fokker-planck --variant dtau-sweep`

Tests

Reduced χ^2 of the co-sim histogram of $\bar{\rho}(\tau = 30)$ against Eq 4. `test3-cosim-matches-fokker-planck · primary · derived_scalar` Sampling noise alone gives ≈ 1 (the 99th percentile of χ^2/k for k ≈ 20–40 bins is about 1.7–1.9).	`range undefined`	pending
Reduced χ^2 of native VCell's histogram of $\bar{\rho}(\tau = 30)$ against Eq 4. `test3-native-matches-fokker-planck · primary · derived_scalar` As above.	`range undefined`	pending
Reduced χ^2 of the co-sim histogram of $\rho(x_0)$ ($\tau = 1$) against the extrapolated Eq T2.1. `test4-cosim-matches-fokker-planck · primary · derived_scalar` As above. The paper reports a relative L2 of 1.9% (i = 0) and 3.3% (i = 1).	`range undefined`	pending
Reduced χ^2 of the co-sim histogram of $\rho(x_0)$ ($\tau = 1$) against the extrapolated Eq T2.1, Test 5 three-cell regime ($d = 50$, $\alpha = 20$, $\beta = 0.5$). `test5-cosim-matches-fokker-planck · primary · derived_scalar` As above. The paper reports 1.3–1.5%.	`range undefined`	pending
Offset of the Test 3 mean-bias fit (offset + slope·Δτ over Δτ = 0.008, 0.004, 0.002, 0.001). Zero means the gap from Eq 4's steady state is a time-discretization error that vanishes as Δτ → 0. `test3-bias-is-first-order-in-dtau · secondary · derived_scalar` A first-order scheme has no Δτ-independent bias. The bound is about 3.7 SE (SE ≈ 0.027%).	`range undefined`	pending

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision

Add a note...

Synthesis & gaps

Where results agree, conflict, or leave open claims — and the priority follow-ups that would resolve them.

Contradictions & uncertainty

NEGATIVE RESULT **— Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions**

Results (results/metrics.json, 2026-10-05; relative L2 = RMS over bins / reference maximum): | test | solver, trials | rel.

Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions

REFUTED

FINDING CONFLICT **— Schaff et al. 2016, Test 2 / Fig 2: coupled calcium sparks in the fast-diffusion limit, both solvers**

Open observation (not a gated test): both solvers show the same small marginal minimum KS p against the well-mixed reference — $7.70 \cdot 10^{-4}$ (co-sim) and $6.87 \cdot 10^{-4}$ (native).

Schaff et al. 2016, Test 2 / Fig 2: coupled calcium sparks in the fast-diffusion limit, both solvers

SUPPORTED

FINDING CONFLICT **— Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions**

DIVERGENCE (failed primary): in the fast-diffusion regime (Test 3) the co-simulation's reduced χ^2 against Eq 4 is 2.48, just over the [0, 2] band, so test3-cosim-matches-fokker-planck FAILS.

Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions

REFUTED

FINDING CONFLICT **— Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions**

The Test 3 gap is a first-order time-discretization error, not a model error: the mean-bias fit offset + slope·Δτ gives a slope of −0.51 ± 0.06 and an offset of −0.044% ± 0.027%, consistent with zero (passes [−0.001, 0.001]).

Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions

REFUTED

CAUSAL CONTINUITY

These studies take their inputs from outside the chain (a supplied feed or reset state) rather than from an upstream study's output — so their result does not yet rest on a modeled upstream mechanism:

Schaff et al. 2016, Test 1 / Fig 1B: separable calcium sparks on both solvers

Validation & reproducibility

Automated, code-computed consistency checks (no AI): conservation/invariant ledgers, computed-vs-authored verdicts, run reproducibility (environment + determinism), parameter drift and acceptance-criterion coverage. Each study's own ledger predicates (e.g. mass/charge balance) are its gating behavior tests, listed on the study's card (its Tests).

Diagnostic study needed `single-channel-fokker-planck`

Study 'single-channel-fokker-planck' verdict is 'failed' with no diagnostic study seeded

Stale finding `F-01`

Finding 'F-01' declared a next_action but no study was seeded

Stale finding `F-01`

Finding 'F-01' declared a next_action but no study was seeded

Stale finding `F-01`

Finding 'F-01' declared a next_action but no study was seeded

Stale finding `F-02`

Finding 'F-02' declared a next_action but no study was seeded

Stale finding `F-02`

Finding 'F-02' declared a next_action but no study was seeded

Stale finding `F-02`

Finding 'F-02' declared a next_action but no study was seeded

Stale finding `F-03`

Finding 'F-03' declared a next_action but no study was seeded

Stale finding `F-03`

Finding 'F-03' declared a next_action but no study was seeded

Stale finding `F-03`

Finding 'F-03' declared a next_action but no study was seeded

Stale finding `F-04`

Finding 'F-04' declared a next_action but no study was seeded

▼ REPORT READINESS — 122 FINDINGS ACROSS 11 CHECKS (0 ERRORS, 122 WARNINGS)

44 **band test missing cites** warning

`cosim-boundary-transfer, dumbbell-three-ways, fenicsx-same-problem, handoff-units-and-binning +8`

18 **runs without emitter** warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +14`

18 **unresolved composite** warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +14`

14 **dag edges legacy only** warning

`bimolecular-hybrid, cosim-boundary-transfer, coupled-sparks-fast-diffusion, coupling-interval-convergence +10`

6 **authored confidence diverges** warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, single-channel-fokker-planck +2`

6 **investigation narrative spine required** warning

`cosim-vs-embedded-hybrid, hybrid-generalizability, vcell-hybrid-paper-benchmarks`

5 **baseline config carry** warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, single-channel-fokker-planck +1`

5 reviewer clarity ambiguity warning

`cosim-boundary-transfer, dumbbell-three-ways, near-membrane-fields, sphere-cosim-vs-native +1`

3 investigation missing adversarial warning

`cosim-vs-embedded-hybrid, hybrid-generalizability, vcell-hybrid-paper-benchmarks`

2 unearned unit labels warning

`near-membrane-fields, sphere-cosim-vs-native`

1 dag edges legacy redundant warning

`field-modulated-decay`

Evidence register

Every study in the campaign (3) — scientific verdict, run status, and one-line result, each linked to its full study card above.

STUDY	Verdict	Run	Result
Schaff et al. 2016, Test 1 / Fig 1B: separable calcium sparks on both solvers	SUPPORTED	ran	Both solvers reproduce Fig 1B's Monte-Carlo convergence: the log-log slope of ε vs N is −0.47 (co-simulation) and −0.46 (native VCell hybrid), both within sampling of the theoretical −1/2.
Schaff et al. 2016, Test 2 / Fig 2: coupled calcium sparks in the fast-diffusion limit, both solvers	SUPPORTED	ran	In the fast-diffusion limit (D = 1000 μm²/s) both hybrid solvers match the exact well-mixed process in mean: max |z| over t∈{1,2,3}s and {U,n} is 1.57 (co-sim) and 0.64 (native VCell), both well under the |z| < 4 thresho
Schaff et al. 2016, Tests 3–5 / Figs 4–6: a coupled single channel against direct Fokker–Planck solutions	REFUTED	ran	Finite diffusion (Tests 4–5): the co-simulation matches the direct Fokker–Planck solutions to within sampling noise in every cell — reduced χ^2 = 0.54 (Test 4, x_0), 0.80 (Test 4, x_1) and 0.80 / 0.89 / 0.86 (Test 5, x_0/x_1/

References

The papers behind each acceptance band, finding and literature anchor.

Schaff 2016 `schaff2016hybrid` — Numerical Approach to Spatial Deterministic-Stochastic Models Arising in Cell Biology · Schaff, James C. and Gao, Fei and Li, Ye and Novak, Igor L. and Slepchenko, Boris M., 2016 · source ↗ 14×

Leave an overall note, and per-study notes on each study card. Then **Review & export** assembles them into a YAML feedback report.

FEEDBACK · THE WHOLE REPORT 👍 Looks right ✏️ Needs revision

Add a note...

Rendered from workspace `viva-pde-particle--outcomes` · deterministic from investigation + study YAML.

`investigations/vcell-hybrid-paper-benchmarks/investigation.yaml · studies/*/study.yaml · loop-trajectory JSON`